

Pre-Automode Report: $D_{s1} \rightarrow D_s \gamma$ LCSR Setup

Ds gamma project

July 22, 2026

1 Purpose

This report summarizes the choices fixed before the autonomous calculation of the radiative transitions

$$D_{s1}(2460) \rightarrow D_s \gamma, \quad D_{s1}(2536) \rightarrow D_s \gamma.$$

The goal of the next phase is to compute the form factor G and the decay width

$$\Gamma(1^+ \rightarrow 0^- \gamma) = \frac{\alpha_{\text{em}}}{3} G^2 |\vec{q}_\gamma|^3, \quad |\vec{q}_\gamma| = \frac{m_{D_{s1}}^2 - m_{D_s}^2}{2m_{D_{s1}}}.$$

2 Fixed Conventions

We use the mostly-minus metric, $g_{\mu\nu} = \text{diag}(+1, -1, -1, -1)$, and

$$\sigma_{\mu\nu} = \frac{i}{2} [\gamma_\mu, \gamma_\nu], \quad \text{Tr}[\gamma_5 \gamma_\mu \gamma_\nu \gamma_\alpha \gamma_\beta] = -4i \epsilon_{\mu\nu\alpha\beta}.$$

The momentum assignment is

$$P = p + q,$$

where P is the initial D_{s1} momentum, p is the final D_s momentum, and q is the photon momentum.

The interpolating currents are

$$J_{A\mu}^{(0)}(x) = \bar{s}(x) \gamma_\mu \gamma_5 Q(x), \tag{1}$$

$$J_{B\mu}^{(0)}(x) = \frac{i}{m_Q + m_s} \bar{s}(x) \sigma_{\mu\alpha} P^\alpha \gamma_5 Q(x), \tag{2}$$

$$J_{D_s}(x) = \bar{s}(x) i \gamma_5 Q(x). \tag{3}$$

The tensor current contracts with P^α , not with the final-state momentum p^α .

3 OPE Organization

The OPE is organized as

$$\Pi_\mu^{\text{OPE}} = \Pi_\mu^{\text{hard}} + \Pi_\mu^{\text{soft}}.$$

3.1 Hard photon emission

The hard sector contains photon emission from both quark lines:

$$\Pi_\mu^{\text{hard}} = -iN_c \int d^4x e^{ip \cdot x} \text{Tr}_D \left[\Gamma_\mu S_Q^\gamma(x) i \gamma_5 S_s^0(-x) + \Gamma_\mu S_Q^0(x) i \gamma_5 S_s^\gamma(-x) \right].$$

The minus sign is the closed-fermion-loop sign from contracting both quark flavors. Propagator and electromagnetic-vertex factors are attached separately. Thus both S_Q^γ and S_s^γ are included. The generic electromagnetic background-field correction is

$$S_q^\gamma(x) = -ie_q \int \frac{d^4k}{(2\pi)^4} e^{-ik \cdot x} \int_0^1 du \left[\frac{\not{k} + m_q}{2(m_q^2 - k^2)^2} F^{\alpha\beta}(ux) \sigma_{\alpha\beta} + \frac{ux_\alpha}{m_q^2 - k^2} F^{\alpha\beta}(ux) \gamma_\beta \right].$$

It is used with $(q, m_q, e_q) = (Q, m_Q, e_Q)$ for the heavy line and (s, m_s, e_s) for the strange line.

The base calculation does not use ordinary condensate-expanded quark propagators. In particular, terms such as $\langle \bar{s}s \rangle$, $\langle \bar{s}g_s \sigma G s \rangle$, and $\langle G^2 \rangle$ are not inserted as propagator corrections.

3.2 Soft photon emission

The soft sector uses photon distribution amplitudes (DAs). For two-particle DAs,

$$\Pi_\mu^{\text{soft}, 2p} = -\frac{i}{4} \int d^4x e^{ip \cdot x} \sum_i \text{Tr}_D[\Gamma_\mu S_Q^0(x) i\gamma_5 \Gamma_i] \langle \gamma(q, \varepsilon) | \bar{s}(x) \Gamma_i s(0) | 0 \rangle.$$

Condensate parameters that appear inside photon DA normalizations, such as $\chi \langle \bar{s}s \rangle$, are kept as photon-wavefunction inputs. They are not counted as condensate-expanded propagators.

For three-particle DAs, the background-gluon propagator insertion is needed:

$$S_Q^G(x) = -ig_s \int \frac{d^4k}{(2\pi)^4} e^{-ik \cdot x} \int_0^1 du \left[\frac{\not{k} + m_Q}{2(m_Q^2 - k^2)^2} G^{\alpha\beta}(ux) \sigma_{\alpha\beta} + \frac{ux_\alpha}{m_Q^2 - k^2} G^{\alpha\beta}(ux) \gamma_\beta \right].$$

This is not a gluon-condensate correction; it is the operator insertion needed for matrix elements of the form $\langle \gamma | \bar{s} \Gamma g_s G_{\alpha\beta} s | 0 \rangle$.

4 Staged Plan

Stage 1 Compute a controlled baseline including hard photon emission and two-particle photon DAs. Keep all explicit strange-quark mass terms, including possible m_s^2 contributions.

Stage 2 Add the three-particle photon DA terms generated by the background-gluon insertion S_Q^G .

5 Numerical Inputs Fixed

Input	Choice	Comment
$\chi(1 \text{ GeV})$	$+3.15 \pm 0.30 \text{ GeV}^{-2}$	sign fixed before automode
θ	35.3° central	scan after central run
m_s	retained	include explicit m_s and m_s^2 terms
Photon DAs	<code>photon_da.py</code>	through twist 4

6 Implemented Artifacts

- `../shared/conventions.md`: core conventions and current definitions.
- `inputs_table.csv`: numerical inputs, now with positive $\chi = +3.15 \text{ GeV}^{-2}$.
- `photon_da.py`: photon DA functions for Stage 1 and Stage 2.

- `scripts/step2_current_building_blocks.wl`: FeynCalc checks of Dirac conventions and current vertices.
- `scripts/step2_hard_photon_numerators.wl`: numerator-level hard-photon traces for heavy-line and strange-line emission.
- `notes/Ds1_to_Ds_gamma_LCSR_notes.tex`: living derivation notes compiled to PDF.

7 Next Automode Tasks

1. Derive the Stage 1 invariant amplitude for the chosen E1 Lorentz structure.
2. Implement the Borel-transformed Stage 1 sum rule.
3. Determine reasonable Borel windows and continuum thresholds.
4. Extract central G_A and G_B , then form the physical mixed amplitudes.
5. Compute $\Gamma[D_{s1}(2460) \rightarrow D_s \gamma]$ and $\Gamma[D_{s1}(2536) \rightarrow D_s \gamma]$.
6. Scan the mixing angle θ and selected input uncertainties.
7. Add Stage 2 as a controlled correction.